MILITARY SPECIFICATION

Control, Test Harness C-9154/APM

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for intermediate shop use and shall facilitate the interface and test of the R-1623/APN Radar Receiver, the RT-1028/APN-202 Receiver Transmitter, the MD-817D Modulator, and the RT-763D/APN-154(V) Receiver Transmitter.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
C-9154/APM	3.5.1

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

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MIL-T-18303	Test Procedures, Pre-production and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS

Military

MIL-STD-461	Electro Magnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definitions and Systems of Units, Electromagnetic Interference Technology.

2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot. Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

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3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.3.1).

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-T-21200.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of MIL-T-21200 for design, construction, and workmanship except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 45 pounds.

3.3.2 Reliability -

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3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 300 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be specified by the contractor.

3.3.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted such that the connecting cables will not interfere with the operating controls. Permanently stamped numbered wires may be used in lieu of color coding.

3.3.3.1 Cable Assemblies, Power Electrical - Applicable input power cable and connections are listed in 3.5.1.7.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in MIL-T-21200.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461 and MIL-STD-462.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200 Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

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3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (30°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input Power Voltage	115 ±5 VAC 400 ±10 Hz
Input DC Voltage	27 ±1 VDC as supplied from the associated equipment Control, Test Harness C-6690A/APM-231 listed in 6.8 or an auxiliary supply as selected by dc power switch.

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or combination of these conditions as specified in MIL-T-21200 for Class 3 equipment.

3.3.9.1 Shock - The equipment shall operate satisfactorily after being subjected to 20 g shock pulses per MIL-T-21200.

3.3.10 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 10 minutes under standard conditions and 20 minutes at extreme service conditions.

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3.3.11 Primary Input Power Requirements - The equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in Specification MIL-T-21200. The power required shall be of the following type and shall not exceed the specified amounts.

DC Power, 28 Volts, 1.5 Amperes (normally supplied by the C-6690A/APM-231)

AC Power, Single Phase, 115V, 60 Hz or 400 Hz, 55 VA

3.4 Performance - The performance of the equipment shall be as set forth in paragraph 3.5. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

3.4.1 Support Equipment Performance Characteristics - The performance characteristics specified herein are those necessary to provide the capability required for the intended usage.

3.4.1.1 Beacon Test, RT-1028 and RT-763D - The equipment shall operate in the "X" band of frequencies and shall be capable of the following characteristics.

3.4.1.1.1 Beacon Interrogation Test Signals - The receiver test signal shall be a pulsed cw signal either single or double-pulse pair of variable spacing.

Frequency	8500 to 9600 MHz
Rise Time	0.1 usec max.
Fall Time	0.2 usec max.
Width	0.35 ± 0.05 usec
Power Range	0 to -70 dBm
Repetition Rate	0 to 10 kHz
Pulse Pair Spacing	1 to 20 usec

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3.4.1.1.2 Beacon Reply Monitor Capabilities - The transmitter reply shall be a pulsed cw signal either single or double-pulse pair of variable spacing. The equipment shall be capable of monitoring the signal with the following characteristics:

Frequency	8500 to 9600 MHz
Rise Time	0.1 usec max.
Fall Time	0.2 usec max.
Width	0.2 to 0.7 usec
Power Range	400 watts peak typical
Repetition Rate	0 to 10 kHz
Pulse Pair Spacing	40 to 165 usec
Beacon Delay	1.0 usec typical

3.4.1.2 Receiver Test, R-1623 - The receiver operates in the K_a-band of frequencies. The equipment shall be capable of the following signal characteristics:

3.4.1.2.1 Receiver Test Signal - The receiver test signal shall be a pulsed cw signal, with or without amplitude modulation, with the following characteristics:

Frequency	33.2 ±2 GHz
Rise Time	50 nanosec max.
Fall Time	100 nanosec max.
Width	200 ±50 nanosec
Power Range	+30 to -40 dBm
Repetition Rate	0 to 2500 kHz
Modulation (AM)	3.5 ±2 dB

3.4.1.2.2 Receiver Output Monitor - The output of the R-1623/APN Receiver shall be an output trigger video pulse for each received rf pulse. Characteristics of the receiver outputs are as follows:

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Pulse Amplitudes	9 volts min.
Rise Time	50 nanosec max.
Pulse Width	0.6 ±0.3 us
Repetition Rate	2000 pps
System Delay	1500 ±10 nanosec
AGC Voltage Range	4.1 to 8.8 VDC
Modulation Frequency	100 Hz

3.4.1.3 Bench Test Parameter Accuracies - The Test Harness, with associated equipment shall be capable of measuring the following parameters to the indicated accuracy.

Parameter	Accuracy
a. Receiver Frequency	±0.03%
b. Receiver Sensitivity	±2.5 dB
c. Spurious Triggering	±1 count
d. Decoding Capability	±0.03 usec
e. Transmitter frequency	±0.03%
f. Transmitter Power	±1 dB
g. Pulse Characteristics	±0.02 usec
h. Bandwidth	±1 MHz
i. Beacon Delay	±0.02 usec
j. Receiver Delay	±0.015 usec
k. DC Voltages	+0.05%
l. Pulse Amplitudes	±5%

3.5 Detail Requirements -

3.5.1 Control, Test Harness C-9154/APM - The harness shall meet the following requirements:

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3.5.1.1 Function - The harness shall provide for interfacing the RT-1028/APN-202 Receiver Transmitter with the C-6690A/APM-231 Control, Test Harness and common ground support equipment (CGSE). It shall include the necessary control switching and interconnections for performance of operational tests. It shall be capable of testing the RT-1028/APN-202 Receiver Transmitter with or without the C-8752/APN-202 Radar Beacon Control.

The harness shall provide for interfacing the R-1623/APM-231 Control, Test Harness and the SM-652/APM Radar Beacon Simulator. It shall include the necessary control switching and interconnections required to conduct operational tests. The harness shall be capable of testing the R-1623/APM Receiver with or without components of the AN/APN-154(V) or the AN/APN-202. Interconnection between the R-1623/APM and components of the AN/APN-154(V) or AN/APN-202 shall be provided, as well as the necessary control functions, so that an ACLS system (less antennas) may be tested.

The harness shall provide for interfacing with the common ground support equipment and shall include the necessary control functions and interconnections for conducting operational tests in the ACLS mode.

The harness shall provide for interfacing the X-band MD-817/APN Modulator with the C-6690A/APM-231 Test Harness and CGSE, and shall include the necessary control functions and interconnections for the performance of tests.

3.5.1.2 Form Factor - The Control, Test Harness C-9154/APM shall be mounted in a carrying case whose maximum case dimensions shall be 23.5 inches long by 10.5 inches high by 16.5 inches wide.

3.5.1.3 Weight - The weight of the equipment shall not exceed 45 pounds.

3.5.1.4 Content - The harness shall provide the following functions.

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3.5.1.4.1 Delay Generator - A delay generator shall be provided to facilitate checking of the delay stabilization characteristics of the R-1623/APN Receiver. It shall have three preset delays that are selectable by means of a front panel control. The characteristics of the delay generator shall be as follows:

Input Trigger

Amplitude	+10 ±1 volts peak
Pulse Width	0.6 ±.2 microseconds
Rise Time	0.1 microseconds max

Output Pulse

Amplitude	-0.6 ±.1 volts peak
Pulse Width	0.3 ±.04 microseconds
Rise Time	60 nanoseconds max

Preset Delays

Low	0.65 ±.05 microseconds
Nominal	0.85 ±.05 microseconds
High	1.05 ±.06 microseconds

3.5.1.4.2 Beacon Control - The equipment shall provide circuitry functionally equivalent to that of the C-8752/APN-202 Radar Beacon Control. Use of this internal control may be selected by the control panel select switch, 3.5.1.5(e).

3.5.1.4.3 Current Monitor - The equipment shall provide separate means for monitoring current input to the R-1623/APN and the RT-1028/APN-202. When selected, the current monitors shall provide an output voltage of 100 millivolts per ampere with one percent accuracy at the digital voltmeter terminals, J18 and J19.

3.5.1.4.4 Modulation Source - A means of supplying current into the amplitude modulation input of the MD-817/APN Modulator or the AM input to the RT-1028/APN-202 Receiver-Transmitter shall be provided in the beacon-only test modes as follows:

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OFF	Modulator terminated to ground through 15,000 ohm ±10%
1.	0.215 ma., dc ±5% short circuit current. Source resistance 60K ohms minimum
2.	3.08 ma., dc ±5% short circuit current. Source resistance 4,000 ohms minimum.
3.	10.2 ma., dc ±5% short circuit current. Source resistance 1200 ohms minimum

3.5.1.4.5 Internal Power Supply - The equipment shall provide a dc voltage suitable for operation of the SM-652/APM Radar Beacon Simulator and as a power source for the equipment internal circuits. It shall have the following characteristics:

Input Voltage	115 ±10 VAC 60 ±10 Hz or 115 ±10 VAC 400 ±20 Hz
Output Voltage	27 ±2 VDC
Output Current	0.35 ampere nominal with 0.75 ampere capability
AC ripple	60 millivolts max. (resistive load)
Regulation (load)	±0.5%
Protection	Primary fused no permanent damage shall occur from an accidental short circuit of the output.

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3.5.1.4.6 RF Detector - The equipment shall provide an rf detector to observe the K_a-band pulse charateristics of the SM-652/APM Radar Simulator. Isolation to the main rf signal shall be provided by a 20 dB waveguide coupler. Characteristics of the rf detector shall be as follows:

Operating Frequency	33.2 ±.4 GHz
Sensitivity at +10 dBm signal	0.2 volts min.
Operating Signal Level	+13 dBm max.

3.5.1.4.7 Panel Light Supply - The equipment shall provide a means for illuminating the lamps of the edge-lighted C-8752/APN and the C-4419B/APN-154(V) control panel. At a nominal load of 0.5 amp amperes the output voltage shall be 5 ±0.5 volts rms with a primary input of 115 Vac.

3.5.1.5 Controls - The following controls shall be provided on the front panel.

(a) MODE - A six position rotary switch, part of the internal C-8752/APN-202 control panel (3.5.1.4.2) used to control the RT-1028/APN-202 operating mode. The following positions are required: OFF, STBY, BCN, TEST, ACLS, XDAT

(b) ENCODE - A six-position rotary switch part of the internal C-8752/APN-202 control panel (3.5.1.4.2) used to select the RT-1028/APN-202 encoder control voltage. The following positions are required: S, 1, 2, 3, 4, and 5.

(c) DECODE - A six-position rotary switch used to select the control voltage applied to the RT-1028/APN-202 decoder control line, part of the internal C-8752/APN-202 control panel. The following positions are required: S, 1, 2, 3, 4, and 5.

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(d) DVM SEL - A 24-position rotary switch used to select the voltage to be measured at the digital voltmeter terminals (J18 and J19). Positions (nine) required for the test of a C-8752/APN-202 panel are as follows: BCN +28V, STANDBY, BCN BIT, SGL/DBL. ACLS BIT. ACLS +28, EXT DATA, DCDR CONT, and ENCDR CONT. Positions (five) required for the test of a C-44195B/APN-154(V) control panel are as follows. BCN +28V, STANDBY, BIT ENB, SGL/DBL, DCDR CONT. Positions (two) required for the test of a R-1623/APN are as follows: INPUT CUR, and AGC. The eight positions required for test of a RT-1028/APN-202 are as follows: INPUT, CUR, PWR SPLY, VIDEO, DCDR, MOD, BCN +20V, DCDR RTN, and ENCDR RTN.

(e) CONT PANEL SEL - A five-position rotary switch used to select the control panel and its mode of operation. This selector permits the testing of, or operation with, external control panels C-8752/APN-202 or C-4419B/APN-154(V). It also permits use of the internal control panel. The following positions are required: INTL C-8752, EXT C-8752, C-4419B, C-4419B TEST, and C-8752 TEST

(f) AM TEST - A four-position rotary switch that selects the current applied to the AM modulator of the beacon under test (3.5.1 4.4). The following positions are required: OFF, 1, 2, 3.

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(g) TEST MODE - An eight-position rotary switch used to select the equipment test mode. The switch controls power to J9 and J7; controls the routing of the transmitter trigger pulses of J5 and J20; controls the input and output of the delay generator (3.5.1.4.1); controls the routing and termination of the inputs to J11: and controls the routing of the amplitude modulation signal and AM source at J7, J12, J16, J17. The following positions are required: RCVR PTCT, RT-1028, R-1623, RT-763D RT-763D R-1623 ' RT-1028 R-1623 ', SPARE, SPARE.

(h) DELAY - A three-position switch for selecting the source of 28 Vdc used to power the RT-1028/APN-202 or R-1623/APN under test. The source may come from the C-6690A/APM-231 or an external supply. The following positions are required. EXT, OFF APM231

(j) AC PWR - On off switch for the 115 Vac power input used to power the internal power supplies. The following positions are required ON, OFF.

(k) FAULT TEST - A two-position switch used to introduce a simulated fault in the RT-1028/APN-202 standby control line to permit testing of the beacon BIT circuits. The two positions required are ON and OFF.

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3.5.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	MS3112E-16-26S	C-8752, connection to external C-8752/APN-202 Control Panel
J2	MS3112E-14-19SW	C-4419B, connections to external C-4119B/APN-154(V) Control Panel
J3	MS3112E-14-19P	CONT PANEL, power and control inputs from C-6690A/APM-231 Control.
J4	MS3112E-16-8P	BCN PWR, RT-763D beacon power and signal inputs from C-6690A/APM-231 Control.
J5	MS3112E-10-6S	RT-763D, PWR, power and signal lines to RT-763D/APN-154(V) Receiver-Transmitter
J6	MS3112E-14-19S	RT-1028 PWR, power and control lines to RT-1028/APN-202 Receiver-Transmitter.
J7	DM5605-27-13S-4-013	R-1623 PWR, power and signal lines to RT-1623/APN Receiver.

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Reference Designation	Receptacle Type	Function
J8	M39012/18-0004	Ka SAMPLE, detected rf output sample from the rf detector (3.5.1.4.6).
J9	MS3112E-10-6S	KA GEN, power output to SM-652/APM Radar Simulator.
J10	MS3112-12-3P	AC PWR, ac power input.
J11	M39012/21-0002	DET RF, detected rf input from beacon.
J12	M39012/21-0002	SPIN ERROR, R-1623/APN Receiver spin error monitor
J13	M39012/21-0002	TRIG MON, RT-1028/APN-202 transmitter trigger monitor
J14	M39012/21-0002	TRIG OUT, transmitter trigger output to RT-1028/APN-202 Receiver-Transmitter.
J15	M39012/21-0002	Beacon detected rf monitor.
J16	M39012/21-0002	EXT AM MON, beacon external amplitude modulation monitor.
J17	M39012/21-0002	EXT AM, external amplitude modulation input to beacon.

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Reference Designation	Receptacle Type	Function
J18	M39024/10-02	+, positive output to digital voltmeter.
J19	M39024/10-2	-, negative output connection for a digital voltmeter.
J20	M39012/21-0002	EXT TRIG, R-1623/APN trigger output monitor.
J21	MS3112E-8-4P	EXT SPLY, dc power input for an external supply
J22	M39012/21-0002	PRE TRIG, pre-trigger output monitor of RT-1028/APN-202 Receiver-Transmitter.
J23	M39012/21-0002	VIDEO TEST, video test point monitor of RT-1028/APN-202
W1	UG-600/U	RF IN, input from SM-652 radar simulator.
W2	UG-600/U	RF OUT, rf output to R-1623/APN.

3.5.1.7 Cable Assemblies - Cable assemblies shall be provided as follows:

3.5.1.7.1 Cable Designation and Length - Cable designation, length, and function shall be as shown below.

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Cable Designation	Nominal Length (inches)	Function
W1	36	Input dc power and control lines from J6 of Test Harness Control C-6990A/APM-231.
W2	42	Beacon dc power input from J1 of Control C-6690A/APM-231.
W3	24	Beacon control lines from J4 of Control C-6690A/APM-231 to RT-763D/APN-154(V) beacon.
W4	40	Beacon dc power from Control C-9154/APM to RT-763D/APM-154(V)
W5	28	Power and signal lines to R-1623/APN Receiver.
W6	72	Input dc power cable from auxiliary power supply.
W7	72	Input ac power cable from primary source
W8	40	Power and control lines to RT-1028 APN-202 Receiver-Transmitter
W9	14	Power and control lines to C-8752/APN-202 Radar Beacon Control.
W10	8	Power and control lines to C-4419B/APN-154(V) Control Panel.

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Cable Designation	Nominal Length (inches)	Function
W11	49	General-purpose trigger and video coax cables to the RT-1028/APN-202 Receiver Transmitter, four required.
None	72	General-purpose video and modulation coax cables to the MD-817/APN Modulator and general usage, three required.

3.5.1.7.2 Cable Connectors - Cables listed in 3.5.1.7.1 shall have the following connectors or their equivalents:

Cable Designation	P1	P2
W1	MS3116F-14-19S	Bendix PT07P014-18P
W2	MS3116F-16-8S	MS3116F-16-8P
W3	MC11E-10-6SW	MS3116F-16-8PW
W4	MS3116F-10-6P	MC11E-10-6SN
W5	DM9700-27-13P-4-003	DM9702-27-13S-4
W6	MS3116F-8-4S	Wires stripped and tinned 0.5 inch nominal.
W7	MS3116F-12-3S	Wires stripped and tinned 0.5 inch.
W8	MS3116-F-14-19P	MS3116F-14-19S
W9	MS3116F-16-26P	(Split harness with two connectors) P2 ... MS3116F-14-19S P3 ... MS3116F-10-6P

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Cable Designation	P1	P2
W10	MS3116F-14-19PW	Bendix PT06E-14-18S (SR)
W11	M39012/16-0001	GRFF 4000-0002 or Gremar 6000 (TNC).
General	M39012/16-0001	M39012/16-0001

3.5.1.8 Control Panel Mount - The equipment shall supply a means for mounting a C-8752/APN-202 or a C-44198B/APN-154(V) control during test.

3.6 Design Data - No data is required by this specification or by applicable documents, unless specified in the contract or order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.4. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors

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not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction test shall include environmental and interference tests in accordance with the requirements of MIL-T-5422, MIL-STD-461, and MIL-STD-462 to the limits specified in MIL-T-21200.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of the production equipment prior to the approval of the preproduction sample is at the contractor's own risk.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the test sample. Evidence of non-compliance with the above shall constitute cause for rejection. For equipment already accepted by the government, it shall be the obligation of the contractor to make necessary corrections as approved be the procuring activity.

Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved test sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

Initial production sample approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following.

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests.

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-in Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.2 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only these adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

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Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4 3 2 Sampling Tests - Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0
Next 50	1
Next 75	1
Next 1000	1
	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

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(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, individual, and special tests.

(3) Manufacturing run-in test specified in paragraph 4.3.1.3 except that the test duration shall be 120 hours minimum with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.3.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.3.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after a significant engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.3.4.2 Scope of Tests - Special tests shall consist of such tests as proved by the procuring activity. Test procedured previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests

4.3.5 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

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(3) Submit to the procuring activity for approval the proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or to require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification os such procedures, including additional tests, when deemed necessary.

4.5 Reconditioning of Tested Equipment - Equipment which has been subjected to sampling or special tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The equipment covered by this specification, in conjunction with the associated equipment of 6.8, is intended to provide the necessary dc power, controls, and interconnections required to facilitate the interface and test of components of the AN/APN-202 and AN/APN-154(V). The applicable conponents of these equipments are listed below:

Component		Type Designation
(a)	Radar Beacon Control	C-8752/APN-202
(b)	Radar Receiver Transmitter	RT-1028/APN-202
(c)	Receiver, Radar	R-1623/APN
(d)	Control	C-4419B/APN-154(V)
(e)	Receiver Transmitter	RT-763D/APN-154(V)
(f)	Modulator, Beacon	MD-817/APN
(g)	Duplexer-circulator	CU-1104/APN-154(V)

6.2 Performance Characteristics - Nominal performance characteristics shall be determined at input voltages of 28 ±0.3 Vdc and 115 ±1.0 Vac as applicable (see 3.3.8). Limiting performance characteristics shall be determined at the highest and lowest values of specified input voltages. These characteristics shall be used when testing the equipment for compliance with the minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or

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reliability will result from the use of materials, parts, and processes other than those specified in Specification MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3.1 Nonrepairable Subassemblies - As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplied). the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 nours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 The parenthesis (*), when used in the type designation. will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The complete type number shall be used on nameplates. shipping records. and instruction books. as applicable.

6.6 Ordering Date - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

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(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.6).

6.7 Asterisk - In specification revisions and superseding amendments, an "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only, and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

(1) AN/APM-231A and associated equipment.

(2) Simulator, Radar Beacon SM-652/APM.

(3) Pulse Generator, HP-214A (two required).

(4) Oscilloscope, AN/USM-281A.

(5) Thermistor Mount, HP-R486A.

(6) Power Meter, HP-431C.

(7) Variable Attenuator, HP-R382A (two required).

(8) Power meter, HP-430C.

(9) Digital Voltmeter, HP-3439A.

(10) Ohmmeter, AN/PSM-4D.

(11) Power Supply, HP-712B.

(12) Counter, AN/USM-207.

(13) AC/DC Range Unit, HP-3445A.

(14) Termination, BNC, 50 ohm (three required).

(15) Tee Adaptor, BNC, UG-274 B/U (three required).

(16) Time Mark Generator, Tektronix 184A.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-81925(AS) CONTROL, TEST HARNESS C-9154/APM

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO (If "yes" in what way?)

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.*)

SUBMITTED BY (*Printed or typed name and activity - Optional*)

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

S/N-0102-014-1801 C-25254

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